Matrix Algebra and the Gauss–Markov Framework

A Crash Course from First Principles

Linear Statistical Models — Lectures 5–6 (3 Aug – 5 Aug)

The second of four crash courses. Crash Course 1 fitted a line. Here we do the general model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where $\mathbf{X}$ may be *rank deficient* — which is exactly the situation in ANOVA. All the linear algebra you need is built up first, from a starting point of vectors, matrices and subspaces.

Contents

1 Why this exists

In simple linear regression, $\mathbf{X}^T\mathbf{X}$ was invertible and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$ was a formula you could just write down. But consider the very first ANOVA-type model:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \qquad i = 1, \ldots, r.$$

Add c to μ and subtract c from every α_i: the model is unchanged. The parameters are **not identifiable**, $\mathbf{X}$ does not have full column rank, and $(\mathbf{X}^T\mathbf{X})^{-1}$ *does not exist.* Everything in Crash Course 1 breaks.

The two questions this course answers

1. If $\boldsymbol{\beta}$ itself cannot be estimated, *what can*?
   Answer: exactly those $a^T\boldsymbol{\beta}$ with $a \in \mathcal{R}(\mathbf{X})$ — the row space.
2. Among all unbiased estimators that are linear in $\mathbf{Y}$, which is best?
   Answer: $a^T\hat{\boldsymbol{\beta}}$, for *any* solution $\hat{\boldsymbol{\beta}}$ of the normal equations. That is the Gauss–Markov

theorem.

2 The linear algebra toolkit

2.1 Column space, row space, rank

For an $n \times p$ matrix A:

$$\mathcal{C}(A) := \{Av : v \in \mathbb{R}^p\} \subseteq \mathbb{R}^n \quad \text{(column space)}, \qquad \mathcal{N}(A) := \{v : Av = 0\} \quad \text{(null space)}.$$

Write $\mathcal{R}(A) := \mathcal{C}(A^T)$ for the row space. The rank $\rho(A)$ is $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$.

Two facts used constantly:

- $\mathcal{C}(A^T) = \mathcal{C}(A^T A)$, hence also $\rho(A) = \rho(A^T A)$.
- $Au = b$ has a solution $\iff$ $b \in \mathcal{C}(A)$.

2.2 Projections

Let U_1, U_2 be subspaces of $\mathbb{R}^n$.

$$U_1 + U_2 = \{u_1 + u_2\}, \qquad U_1 \oplus U_2 = U_1 + U_2 \text{ when the representation } u = u_1 + u_2 \text{ is unique.}$$

Uniqueness holds iff $U_1 \cap U_2 = \{0\}$. If $u = u_1 + u_2$ uniquely, the map $P(u) = u_2$ is the **projection onto** U_2 **along** U_1, and it is linear, so $P(u) = Pu$ for a matrix P.

Characterisation of a projection matrix (TFAE)

(i) P is a projection matrix;

(ii) P is **idempotent**, $P^2 = P$;

(iii) $\mathcal{N}(P) = \mathcal{C}(I - P)$;

(iv) $\rho(P) + \rho(I - P) = n$;

(v) $\mathcal{C}(P) + \mathcal{C}(I - P)$ is a direct sum.

Two consequences you will use without thinking:

- Eigenvalues of an idempotent matrix are 0 and 1 only (if $Pv = \lambda v$ then $\lambda v = Pv = P^2 v = \lambda^2 v$, so $\lambda^2 = \lambda$).
- Therefore $\text{tr}(P) = \rho(P)$ — the trace *counts* the dimension of the image. This is why traces keep turning into degrees of freedom.

2.3 Orthogonal projections

Projecting onto S *along* $S^\perp$ gives the **orthogonal** projection.

Characterisation of an orthogonal projector (TFAE)

(i) P is an orthogonal projector;

(ii) $P = P^T P$;

(iii) $P^T = P$ **and** $P^2 = P$ (symmetric *and* idempotent).

Proof. (iii)⇒(ii) is immediate. For (i)⇒(iii): $\mathcal{C}(I - P) = S^\perp$ and $\mathcal{C}(P) = S$ are orthogonal, so $P^T(I - P) = 0$, giving $P^T = P^T P$; the right side is symmetric, so $P^T = P$, and then $P = P^2$. ■

The formula for the orthogonal projector onto $\mathcal{C}(\mathrm{X})$

$$P_{\mathbf{X}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T,$$

where $(\mathbf{X}^T\mathbf{X})^-$ is any symmetric g-inverse (§2.4). When $\mathbf{X}$ has full column rank this is the familiar $\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$. **$P_{\mathbf{X}}$ does not depend on which g-inverse you pick** — see §2.5.

2.4 Generalised inverses

Definition

G is a **g-inverse** of A (written A^-) if

$$AGA = A.$$

Why the definition is exactly right. If $b \in \mathcal{C}(A)$, so $b = Au$ for some u, then $x = Gb$ solves $Ax = b$:

$$A(Gb) = AGAu = Au = b.$$

So a g-inverse is precisely a device for solving consistent linear systems. Nothing more is demanded of it, which is why it is not unique.

Every matrix has one. Take a full-rank factorisation $A_{m\times n} = P_{m\times r}Q^T_{r\times n}$ with $r = \rho(A)$. Then P has a left inverse and Q^T a right inverse, and one assembles G with $AGA = A$.

For symmetric A, build it from the spectrum. Write $A = P\Lambda P^T$ with $PP^T = P^TP = I$, $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$. Try $G = P\Delta P^T$ with $\Delta = \mathrm{diag}(\delta_1, \dots, \delta_n)$. Then

$$AGA = A \iff \Lambda\Delta\Lambda = \Lambda \iff \lambda_i^2\delta_i = \lambda_i \quad \forall i \iff \delta_i = \begin{cases} 1/\lambda_i & \lambda_i \neq 0, \\ \text{anything} & \lambda_i = 0. \end{cases}$$

The freedom sits exactly on the null space. Choosing the "anything" to be 0 gives the **Moore–Penrose inverse**

$$A^+ = P\Delta P^T, \qquad \delta_i = \begin{cases} 1/\lambda_i & \lambda_i \neq 0 \\ 0 & \lambda_i = 0 \end{cases} \qquad \text{i.e.} \quad \Lambda = \begin{pmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \Delta = \begin{pmatrix} \Lambda_1^{-1} & 0 \\ 0 & 0 \end{pmatrix},$$

which yields the solution of *smallest norm*.

2.5 The g-inverse facts worth memorising

Statement	Use
General solution of $Ax = y$ ($y \in \mathcal{C}(A)$) is $A^-y + (I - A^-A)z$, $z \in \mathbb{R}^n$ arbitrary.	Describes *all* solutions of the normal equations.
G is a g-inverse of A $\iff$ AG is idempotent with $\rho(AG) = \rho(A)$ $\iff$ GA likewise.	Lets you verify a candidate.
AG is the projection onto $\mathcal{C}(A)$ along $\mathcal{N}(AG)$.	Connects g-inverses to projections.
$A^T(AA^T)^-$ and $(A^TA)^-A^T$ are always g-inverses of A.	Constructive.
If $\mathcal{R}(A) \subseteq \mathcal{R}(B)$ and $\mathcal{C}(C) \subseteq \mathcal{C}(B)$, then AB^-C is **invariant** to the choice of B^-.	**This is why $P_{\mathbf{X}}$ and $a^T\hat{\boldsymbol{\beta}}$ are well defined.**
A^- is *reflexive* if $A^-AA^- = A^-$; *minimum-norm* if $(A^-A)^T = A^-A$; *least-squares* if $(AA^-)^T = AA^-$. Moore–Penrose satisfies all three.	Vocabulary.

3 The Gauss–Markov model

The setup

$$\underset{n\times 1}{\mathbf{Y}} = \underset{n\times p}{\mathbf{X}}\ \underset{p\times 1}{\boldsymbol{\beta}} + \underset{n\times 1}{\boldsymbol{\varepsilon}}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I_n), \qquad \mathbf{X} \text{ non-random.}$$

Unknowns: $\boldsymbol{\beta}$ and σ^2. No assumption that $\rho(\mathbf{X}) = p$.

3.1 Least squares and the normal equations

$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Differentiating,

$$\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = 0 \quad \iff \quad \mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{Y} \qquad (\textbf{normal equations}).$$

The normal equations *always* have a solution

A system $Au = b$ is solvable iff $b \in \mathcal{C}(A)$. Here $b = \mathbf{X}^T\mathbf{Y} \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) = \mathcal{C}(A)$. So a solution always exists — but it is unique only when $\mathbf{X}^T\mathbf{X}$ is non-singular, i.e. $\rho(\mathbf{X}) = p$.

If $\mathbf{X}^T\mathbf{X}$ is singular and $\hat{\boldsymbol{\beta}}$ is a solution, so is $\hat{\boldsymbol{\beta}} + z$ for every $z \in \mathcal{N}(\mathbf{X}^T\mathbf{X})$. The general solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$ for any g-inverse.

The fitted values, however, are always unique:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = P_{\mathbf{X}}\mathbf{Y}, \qquad \hat{\sigma}^2 = \frac{\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2}{n - \rho(\mathbf{X})}.$$

Common confusion: $n - p$ or $n - \rho(\mathbf{X})$?

The residual degrees of freedom is $n - \rho(\mathbf{X})$, the *rank*, not the number of columns. In full-rank regression they agree. In ANOVA with r groups written as $\mu + \alpha_i$ there are $r + 1$ columns but rank r, so the d.f. is $n - r$ — not $n - r - 1$. Getting this wrong corrupts every MSE, every t, and every F in the question.

3.2 Identifiability

Definition and theorem

A family $\{P_\theta\}$ is **identifiable** if $P_{\theta_1} = P_{\theta_2} \Rightarrow \theta_1 = \theta_2$.
Theorem. The linear model $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$ is identifiable $\iff$ $\mathbf{X}^T\mathbf{X}$ is non-singular.

Proof. $\mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_1, \sigma_1^2 I) = \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_2, \sigma_2^2 I)$ iff $\mathbf{X}\boldsymbol{\beta}_1 = \mathbf{X}\boldsymbol{\beta}_2$ and $\sigma_1^2 = \sigma_2^2$ (a normal law is determined by its mean and variance). So identifiability is exactly the statement "$\mathbf{X}\boldsymbol{\beta}_1 = \mathbf{X}\boldsymbol{\beta}_2 \Rightarrow \boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$", i.e. $\mathcal{N}(\mathbf{X}) = \{0\}$, i.e. $\rho(\mathbf{X}) = p$, i.e. $\mathbf{X}^T\mathbf{X}$ invertible. ∎

4 Estimability — the central idea

Definition

Estimable means: an unbiased estimator exists. A linear parametric function $a^T\boldsymbol{\beta}$ is **linearly estimable** if there is an $\ell \in \mathbb{R}^n$ with $\mathbb{E}(\ell^T\mathbf{Y}) = a^T\boldsymbol{\beta}$ for **all** $\boldsymbol{\beta}$.

The estimability criterion

$$\boxed{a^T\boldsymbol{\beta} \text{ is estimable} \iff a \in \mathcal{R}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}).}$$

Proof. $\ell^T\mathbf{Y}$ is unbiased for $a^T\boldsymbol{\beta} \iff \ell^T\mathbf{X}\boldsymbol{\beta} = a^T\boldsymbol{\beta}$ for all $\boldsymbol{\beta} \iff \ell^T\mathbf{X} = a^T \iff a = \mathbf{X}^T\ell \iff a \in \mathcal{C}(\mathbf{X}^T)$. ■

How to check estimability in an exam

Two equivalent tests. Pick whichever is easier for the matrix in front of you.

1. **Row-space test.** Ask whether a^T is a linear combination of the rows of $\mathbf{X}$. Write $a^T = c^T\mathbf{X}$ and try to solve for c; if you can, it is estimable.
2. **Null-space test.** $a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{N}(\mathbf{X})^\perp$, so

$$a^T\boldsymbol{\beta} \text{ estimable} \iff a^Tz = 0 \text{ for every } z \in \mathcal{N}(\mathbf{X}).$$

   Find a basis of $\mathcal{N}(\mathbf{X})$ once, then testing any a is a dot product. *This is usually the faster route.*

4.1 The worked example from the notes

Take $Y_{ij} = \alpha_i + \gamma_j + \varepsilon_{ij}$, $i, j \in \{1, 2\}$, with $\boldsymbol{\beta} = (\alpha_1, \alpha_2, \gamma_1, \gamma_2)^T$ and rows ordered $Y_{11}, Y_{12}, Y_{21}, Y_{22}$:

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad \rho(\mathbf{X}) = 3.$$

Rank is 3, not 4: the model is not identifiable and $\boldsymbol{\beta}$ cannot be estimated. Writing $a^T = c^T\mathbf{X}$ with $c = (c_1, c_2, c_3, c_4)^T$ gives

$$a^T = (c_1 + c_2,\ c_3 + c_4,\ c_1 + c_3,\ c_2 + c_4),$$

so the constraint on $a = (a_1, a_2, a_3, a_4)^T$ is

$$\boxed{a_1 + a_2 = a_3 + a_4}.$$

Check the null-space route as well: $z = (1, 1, -1, -1)^T$ satisfies $\mathbf{X}z = 0$, and $a^Tz = a_1 + a_2 - a_3 - a_4 = 0$ is the same condition.

Function	a	Estimable?
α_1	$(1, 0, 0, 0)$	**No** $(1 + 0 \neq 0 + 0)$
γ_1	$(0, 0, 1, 0)$	**No**
$\alpha_1 + \gamma_1$	$(1, 0, 1, 0)$	**Yes** $(1 = 1)$
$\gamma_1 - \gamma_2$	$(0, 0, 1, -1)$	**Yes** $(0 = 0)$
$\alpha_1 - \alpha_2$	$(1, -1, 0, 0)$	**Yes** $(0 = 0)$

The moral, which is the moral of ANOVA

Individual effects are not estimable; **differences** of effects and **contrasts** are. This is exactly why in one-way ANOVA you can never estimate α_i on its own but you can always estimate $\alpha_i - \alpha_{i'} = \mu_i - \mu_{i'}$, and more generally $\sum c_i \alpha_i$ whenever $\sum c_i = 0$.

4.2 Uniqueness of the estimate of an estimable function

Let $\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\beta}}_2$ both solve $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{Y}$. Estimability gives $a = \mathbf{X}^T\mathbf{X}c$ for some c, so

$$a^T\hat{\boldsymbol{\beta}}_1 - a^T\hat{\boldsymbol{\beta}}_2 = c^T\mathbf{X}^T\mathbf{X}(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2) = c^T(\mathbf{X}^T\mathbf{Y} - \mathbf{X}^T\mathbf{Y}) = 0.$$

So although $\hat{\boldsymbol{\beta}}$ is not unique, the *estimate of an estimable function* is. This is what makes the whole rank-deficient theory workable.

Also note the unbiasedness: with $a = \mathbf{X}^T\mathbf{X}u$, $a^T\hat{\boldsymbol{\beta}} = u^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = u^T\mathbf{X}^T\mathbf{Y}$, hence $\mathbb{E}(a^T\hat{\boldsymbol{\beta}}) = u^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = a^T\boldsymbol{\beta}$.

5 BLUE, zero functions, and the Gauss–Markov theorem

Definitions

- A **linear unbiased estimator** (LUE) of $a^T\boldsymbol{\beta}$ is any $\ell^T\mathbf{Y}$ with $\mathbb{E}(\ell^T\mathbf{Y}) = a^T\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$.
- A LUE $\ell^T\mathbf{Y}$ is the **BLUE** if $\operatorname{Var}(\ell^T\mathbf{Y}) \le \operatorname{Var}(m^T\mathbf{Y})$ for every LUE $m^T\mathbf{Y}$ of the same $a^T\boldsymbol{\beta}$.
- $c^T\mathbf{Y}$ is a **linear zero function** (LZF) if $\mathbb{E}(c^T\mathbf{Y}) = 0$ for all $\boldsymbol{\beta}$, i.e. $c^T\mathbf{X}\boldsymbol{\beta} = 0\ \forall\boldsymbol{\beta}$, i.e. $\mathbf{X}^Tc = 0$.

Geometrically: LZFs are the linear functions of $\mathbf{Y}$ carrying *no information about* $\boldsymbol{\beta}$ — they are the residual directions, $c \in \mathcal{N}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X})^\perp$.

5.1 The characterisation of a BLUE

Proposition

A LUE $\ell^T\mathbf{Y}$ is a BLUE $\iff$ $\operatorname{Cov}(\ell^T\mathbf{Y}, c^T\mathbf{Y}) = 0$ for *every* LZF $c^T\mathbf{Y}$.

Proof. ($\Leftarrow$) Let $m^T\mathbf{Y}$ be any other LUE. Then $(m-\ell)^T\mathbf{Y}$ has expectation $a^T\boldsymbol{\beta} - a^T\boldsymbol{\beta} = 0$, so it is an LZF. Hence

$$\operatorname{Var}(m^T\mathbf{Y}) = \operatorname{Var}(\ell^T\mathbf{Y} + (m-\ell)^T\mathbf{Y}) = \operatorname{Var}(\ell^T\mathbf{Y}) + \operatorname{Var}((m-\ell)^T\mathbf{Y}) + 2\underbrace{\operatorname{Cov}(\ell^T\mathbf{Y}, (m-\ell)^T\mathbf{Y})}_{=0} \ge \operatorname{Var}(\ell^T\mathbf{Y}).$$

($\Rightarrow$) Let $\ell^T\mathbf{Y}$ be BLUE and $c^T\mathbf{Y}$ an LZF. For every $\lambda \in \mathbb{R}$, $\ell^T\mathbf{Y} + \lambda c^T\mathbf{Y}$ is still a LUE, so

$$\operatorname{Var}(\ell^T\mathbf{Y}) \le \operatorname{Var}(\ell^T\mathbf{Y}) + \lambda^2\operatorname{Var}(c^T\mathbf{Y}) + 2\lambda\operatorname{Cov}(\ell^T\mathbf{Y}, c^T\mathbf{Y}),$$

i.e. $0 \le \lambda^2\operatorname{Var}(c^T\mathbf{Y}) + 2\lambda\operatorname{Cov}(\ell^T\mathbf{Y}, c^T\mathbf{Y})$ for *all* λ. If $\operatorname{Cov} \ne 0$, choose λ small with the opposite sign to the covariance: the linear term dominates the quadratic one and the right side is negative — contradiction. Hence $\operatorname{Cov}(\ell^T\mathbf{Y}, c^T\mathbf{Y}) = 0$. ∎

The picture behind this proof

Decompose any LUE as (the BLUE) + (an LZF). The BLUE lies in $\mathcal{C}(\mathbf{X})$, the LZFs in $\mathcal{C}(\mathbf{X})^{\perp}$. Being uncorrelated with all LZFs means "having no wasted component in the noise directions". **BLUE = orthogonal projection, again.**

5.2 The Gauss–Markov theorem

Gauss–Markov Theorem

In the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\operatorname{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$:
if $a^T\boldsymbol{\beta}$ is estimable, then $a^T\hat{\boldsymbol{\beta}}$ is its **unique BLUE**, where $\hat{\boldsymbol{\beta}}$ is *any* solution of the normal equations $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{Y}$.

Proof. Estimability gives $a^T = \ell^T\mathbf{X}$ for some ℓ, so $a^T\hat{\boldsymbol{\beta}} = \ell^T\mathbf{X}\hat{\boldsymbol{\beta}}$. Any solution has the form $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y} + z$ with $z \in \mathcal{N}(\mathbf{X}^T\mathbf{X})$; but $\mathcal{N}(\mathbf{X}^T\mathbf{X}) = \mathcal{N}(\mathbf{X})$, so $\mathbf{X}z = 0$ and

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y} = P_{\mathbf{X}}\mathbf{Y}.$$

Therefore $a^T\hat{\boldsymbol{\beta}} = \ell^T P_{\mathbf{X}}\mathbf{Y}$, which is linear in $\mathbf{Y}$, and

$$\mathbb{E}(a^T\hat{\boldsymbol{\beta}}) = \ell^T P_{\mathbf{X}}\mathbf{X}\boldsymbol{\beta} = \ell^T\mathbf{X}\boldsymbol{\beta} = a^T\boldsymbol{\beta},$$

so it is a LUE. For optimality, write $a = \mathbf{X}^T\mathbf{X}u$ (possible by estimability), so $a^T\hat{\boldsymbol{\beta}} = u^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = u^T\mathbf{X}^T\mathbf{Y}$. Let $h^T\mathbf{Y}$ be any LZF, so $\mathbf{X}^T h = 0$. Then

$$\operatorname{Cov}(a^T\hat{\boldsymbol{\beta}},\ h^T\mathbf{Y}) = \operatorname{Cov}(u^T\mathbf{X}^T\mathbf{Y}, h^T\mathbf{Y}) = u^T\mathbf{X}^T\operatorname{Var}(\mathbf{Y})h = \sigma^2 u^T\mathbf{X}^T h = 0.$$

By the previous proposition, $a^T\hat{\boldsymbol{\beta}}$ is the BLUE. Uniqueness follows from §4.3. ■

This is Homework 2, Question 1

HW2 Q1 walks exactly this route: (a) $\hat{\boldsymbol{\mu}} = P_{\mathbf{X}}\mathbf{Y}$; (b) $\theta = \mathbf{X}^T\boldsymbol{\mu}$ is estimable with BLUE $\hat{\theta} = \mathbf{X}^T\mathbf{Y}$; (c) if $a^T\theta$ is estimable, its BLUE is $w^T\hat{\theta}$. For (b) note $\theta = \mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$, so $a =$ $\mathbf{X}^T\mathbf{X}$-times-something for each coordinate — estimability is automatic — and the BLUE is $\mathbf{X}^T\hat{\boldsymbol{\mu}} = \mathbf{X}^T P_{\mathbf{X}}\mathbf{Y} = \mathbf{X}^T\mathbf{Y}$ since $\mathbf{X}^T P_{\mathbf{X}} = (P_{\mathbf{X}}\mathbf{X})^T = \mathbf{X}^T$.

6 Generalised least squares: correlated errors

Suppose $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, Σ positive definite. Now LSE and MLE part company.
LSE still solves $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{Y}$ — it does not know about Σ.
MLE maximises $-\frac{1}{2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T\Sigma^{-1}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) - \frac{1}{2}\log|\Sigma|$, so

$$\hat{\boldsymbol{\beta}}^{ML} = \arg\min_{\boldsymbol{\beta}}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T\Sigma^{-1}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \quad \Longrightarrow \quad \mathbf{X}^T\Sigma^{-1}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\Sigma^{-1}\mathbf{Y}.$$

With $\mathbf{X}$ of full column rank,

$$\hat{\boldsymbol{\beta}}^{LS} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}, \qquad \hat{\boldsymbol{\beta}}^{ML} = (\mathbf{X}^T\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}^T\Sigma^{-1}\mathbf{Y}.$$

Both are unbiased, but only one is best

$$\mathrm{Var}(a^T\hat{\boldsymbol\beta}^{LS}) = a^T(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T\Sigma\mathbf{X})(\mathbf{X}^T\mathbf{X})^{-1}a, \qquad \mathrm{Var}(a^T\hat{\boldsymbol\beta}^{ML}) = a^T(\mathbf{X}^T\Sigma^{-1}\mathbf{X})^{-1}a,$$

and the second is always $\leq$ the first. The right version of Gauss–Markov here is: apply the ordinary theorem to the transformed model

$$\Sigma^{-1/2}\mathbf{Y} = \Sigma^{-1/2}\mathbf{X}\boldsymbol\beta + \Sigma^{-1/2}\boldsymbol\varepsilon,$$

whose error has variance I. So the BLUE of an estimable $a^T\boldsymbol\beta$ is $a^T\hat{\boldsymbol\beta}^{ML}$, the **generalised least squares** estimator. Ordinary least squares is BLUE only when $\Sigma = \sigma^2 I$ (or, more precisely, when $\mathcal{C}(\Sigma\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$).

7 Practice, with answers

P1. Show $\mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$ and hence $\mathcal{N}(\mathbf{X}) = \mathcal{N}(\mathbf{X}^T\mathbf{X})$.

P2. In the one-way model $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$ with $r = 3$ groups, find a basis for $\mathcal{N}(\mathbf{X})$ and use it to decide which of μ, α_1, $\alpha_1 - \alpha_2$, $\mu + \alpha_1$, $\alpha_1 + \alpha_2 - 2\alpha_3$ are estimable.

P3. Prove that $P_{\mathbf{X}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T$ does not depend on the choice of g-inverse.

P4. Give an example of an LZF in the simple linear regression model, and verify directly that it is uncorrelated with $\hat\beta_1$.

P5. $\mathbf{Y} \sim \mathcal{N}_3\big((\alpha_1,\ \alpha_1+\alpha_2,\ \alpha_1+\alpha_3)^T,\ \sigma^2 I_3\big)$. Is $\alpha_2 - \alpha_3$ estimable? If so, find its BLUE and its variance.

Answers

P1. $\mathcal{C}(\mathbf{X}^T\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^T)$ is clear. Conversely if $\mathbf{X}^T\mathbf{X}v = 0$ then $v^T\mathbf{X}^T\mathbf{X}v = \|\mathbf{X}v\|^2 = 0$, so $\mathbf{X}v = 0$; hence $\mathcal{N}(\mathbf{X}^T\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X})$, and the reverse inclusion is trivial, so the two null spaces coincide. Equal null spaces of p-column matrices force equal ranks, so $\rho(\mathbf{X}^T\mathbf{X}) = \rho(\mathbf{X}) = \rho(\mathbf{X}^T)$, and the inclusion of column spaces above must be equality.

P2. With $\boldsymbol\beta = (\mu, \alpha_1, \alpha_2, \alpha_3)^T$, every row of $\mathbf{X}$ has the form $(1, e_i^T)$, so $\mathbf{X}z = 0$ forces $z = c(1, -1, -1, -1)^T$. Then $a^T\boldsymbol\beta$ is estimable iff $a_0 - a_1 - a_2 - a_3 = 0$ where $a = (a_0, a_1, a_2, a_3)$: μ (i.e. $a = (1,0,0,0)$) gives $1 \neq 0$, **not estimable**; α_1 gives $-1 \neq 0$, **not estimable**; $\alpha_1 - \alpha_2$ gives 0, **estimable**; $\mu + \alpha_1$ gives $1 - 1 = 0$, **estimable** (it is μ_1); $\alpha_1 + \alpha_2 - 2\alpha_3$ gives $-(1 + 1 - 2) = 0$, **estimable** (a contrast).

P3. Apply the invariance rule: $AB^{-}C$ is invariant to B^{-} when $\mathcal{R}(A) \subseteq \mathcal{R}(B)$ and $\mathcal{C}(C) \subseteq \mathcal{C}(B)$. Take $A = \mathbf{X}$, $B = \mathbf{X}^T\mathbf{X}$, $C = \mathbf{X}^T$. Then $\mathcal{R}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) = \mathcal{R}(B)$ and $\mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) = \mathcal{C}(B)$, so both conditions hold with equality.

P4. Any c with $\mathbf{X}^Tc = 0$, i.e. $\sum c_i = 0$ and $\sum c_i x_i = 0$. For instance with $n = 3$ and $x = (1, 2, 3)$, take $c = (1, -2, 1)$. Then $\mathrm{Cov}(c^T\mathbf{Y}, \hat\beta_1) = \sigma^2 c^T\mathbf{w}$ where $w_i = (x_i - \bar{x})/S_x^2$; since $c^T\mathbf{w} = \frac{1}{S_x^2}(\sum c_i x_i - \bar{x}\sum c_i) = 0$, they are uncorrelated. (More generally $c^T\mathbf{Y} = c^T\boldsymbol\varepsilon$ is a residual direction and $\hat{\boldsymbol\beta}$ lives in the column space.)

P5. Here $\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, which has rank 3 — full rank — so *everything* is estimable and $\hat{\boldsymbol\beta} = \mathbf{X}^{-1}\mathbf{Y}$. Solving: $\hat\alpha_1 = Y_1$, $\hat\alpha_2 = Y_2 - Y_1$, $\hat\alpha_3 = Y_3 - Y_1$. Hence

$$\widehat{\alpha_2 - \alpha_3} = Y_2 - Y_3, \qquad \mathrm{Var} = 2\sigma^2.$$

(Sanity check via $a^T = c^T\mathbf{X}$ with $a = (0, 1, -1)$: $c = (0, 1, -1)$ works, and indeed $c^T\mathbf{Y} = Y_2 - Y_3$.)

Companions: Simple Linear Regression; Quadratic Forms, Nested Models and Testing; One-Way ANOVA.